In [ ]:

```python
#Name: Meenakshi Sangeeth
#Registration No: 240905630
#Section: PG
#Roll No: 34
```

In [ ]:

```python
#lab Manual
```

In [3]:

```python
#1. Write a Python code block that inputs numbers into a list. Print the
n=int(input("Enter the number of elements:"))
l=[]
sum=0
for i in range(n):
    x=int(input("Enter element:"))
    sum+=x
    l.append(x)
print("The list:",l)
l.sort()
print("Largest element:",l[n-1])
print("Smallest element:",l[0])
print("Sum:",sum)
print("Average:",sum/n)
```

```
Enter the number of elements:5
Enter element:2
Enter element:7
Enter element:4
Enter element:8
Enter element:5
The list: [2, 7, 4, 8, 5]
Largest element: 8
Smallest element: 2
Sum: 26
Average: 5.2
```

In [8]:

```python
#2. Write a Python code block to create a tuple with five elements. Try
l=[]
for i in range(5):
    x=int(input("Enter element:"))
    l.append(x)
print(tuple(l))
tup[2]=2
#The above statement will result in an error because tuple is an immutal
```

```
Enter element:1
Enter element:2
Enter element:3
Enter element:4
Enter element:5
(1, 2, 3, 4, 5)

---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_15496\786979261.py in <module>
      5     l.append(x)
      6 print(tuple(l))
----> 7 tup[2]=2

TypeError: 'tuple' object does not support item assignment
```

In [40]:

```python
#3. Write a Python code block to create a dictionary of cricket World Cu

d={}
n=int(input("Enter the number of Cricket World Cup Winners:"))
for i in range(n):
    key=int(input("Enter the year:"))
    val=input("Enter the country:")
    d[key]=val
print(d)
count = {}
for i in d.values():
    if i in count:
        count[i] += 1
    else:
        count[i] = 1

best = max(count,key=count.get)
print("The best performing country is:",best)


unique = list(count.keys())
print("Unique countries:", unique)
```

```
Enter the number of Cricket World Cup Winners:5
Enter the year:2001
Enter the country:india
Enter the year:2002
Enter the country:srilanka
Enter the year:2003
Enter the country:india
Enter the year:2004
Enter the country:australia
Enter the year:2005
Enter the country:india
{2001: 'india', 2002: 'srilanka', 2003: 'india', 2004: 'australia', 2005:
'india'}
The best performing country is: india
Unique countries: ['india', 'srilanka', 'australia']
```

In [42]:

```python
#4.Write a Python code block that inputs a sentence from the user. Count

str=input("Enter a sentence:")
words=str.split()
count={}
for i in words:
    n=words.count(i)
    count[i]=[n]
print(count)
```

```
Enter a sentence:hi hello hi hello hi
{'hi': [3], 'hello': [2]}
```

In [45]:

```python
#5.Write a Python code block to input numbers into two sets.  Perform u
s1=set()
n1=int(input("Enter the number of elements in set 1:"))
for i in range(n1):
    x=int(input("Enter the element:"))
    s1.add(x)
s2=set()
n2=int(input("Enter the number of elements in set 2:"))
for i in range(n2):
    y=int(input("Enter the element:"))
    s2.add(y)

union=s1|s2
intersection=s1&s2
difference=s1-s2
print("Union of the two sets is ",union)
print("Intersection of the two sets is ",intersection)
print("Difference of the two sets is ",difference)
```

```
Enter the number of elements in set 1:3
Enter the element:1
Enter the element:2
Enter the element:3
Enter the number of elements in set 2:3
Enter the element:2
Enter the element:3
Enter the element:4
Union of the two sets is  {1, 2, 3, 4}
Intersection of the two sets is  {2, 3}
Difference of the two sets is  {1}
```

In [ ]:

```python
#Week 2 – Additional Questions
```

In [53]:

```python
#1.Write a Python function to return the longest string from a list of s

l=[]
c=0
n=int(input("Enter the number of strings:"))
for i in range(n):
    s=input("Enter the string:")
    count=len(s)
    if count>c:
        c=count
        longest=s
    l.append(s)
print(l)
print("Longest String:",longest)

```

```
Enter the number of strings:2
Enter the string:hi
Enter the string:hello
['hi', 'hello']
Longest String: hello
```

In [54]:

```python
#2. Write a Python function that takes a list of integers and returns a
l=[]
l1=[]
n=int(input("Enter the number of elements:"))
for i in range(n):
    x=int(input("Enter integer:"))
    l.append(x)
    if x>5:
        l1.append(x)
print(l)
print(l1)
```

```
Enter the number of elements:9
Enter integer:1
Enter integer:2
Enter integer:3
Enter integer:4
Enter integer:5
Enter integer:6
Enter integer:7
Enter integer:8
Enter integer:9
[1, 2, 3, 4, 5, 6, 7, 8, 9]
[6, 7, 8, 9]
```

In [57]:

```python
#3. Write a Python function to swap Two Numbers using a Tuple.
a=int(input("Enter a:"))
b=int(input("Enter b:"))
print("a:",a)
print("b:",b)
(b, a) = (a, b)
print("After swapping:")
print("a:",a)
print("b:",b)
```

```
Enter a:2
Enter b:3
a: 2
b: 3
After swapping:
a: 3
b: 2
```

In [ ]:

```python

```

In [ ]:

```python

```